Unreported competition - Repaired practice edition - Answer key

Division B | Water Quality | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (4 points) Authored written substitute for unavailable source material: Explain the evidence supporting your answer, including the calculation or comparison.

Unit: Calibrate a Division B salinometer.
Worked-case data: 4.0% standard: Immersion is 9.8 cm. 6.0% standard: Immersion is 9.6 cm. Unknown: Immersion is 9.7 cm at the same temperature.
Separate investigation observations:
4.0% standard: Immersion is 9.8 cm.
6.0% standard: Immersion is 9.6 cm.
Unknown: Immersion is 9.7 cm at the same temperature.

answer: The unknown lies halfway between the two nearby calibration depths. Estimate halfway between 4.0% and 6.0%. Report about 5.0% mass/volume; real calibration curves may not be perfectly linear. (4 points).